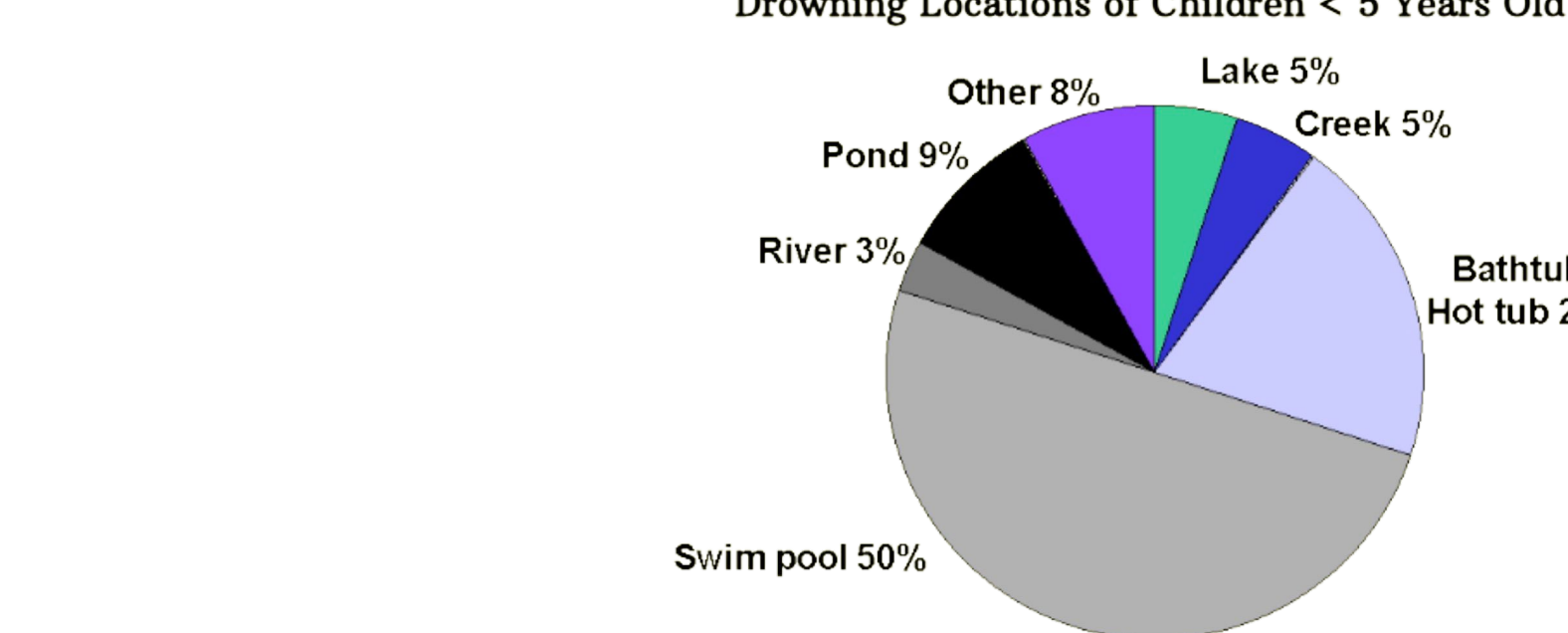


Introduction and Hypothesis

Drowning is the leading cause of death in children aged 1-4 years old. It is necessary to address the root of the problem - automated monitoring of pools. The CDC estimates that around 4,000 fatal and 8,000 nonfatal drownings occur yearly in the US alone and that 75% of these occur in residential pools without supervision. The solution to this problem lies in the detection and prevention of drowning, especially when children are swimming without adult supervision. Currently, children make up a majority of drowning cases as they lack extensive swimming practice and are not always under supervision. From falling into a pool to swimming when unsupervised, a significant amount of time could pass before finding the drowning victim. However, it is well understood that drowning is not an immediate process and takes between 30 to 60 seconds of struggling above the water before submerging below the water, where individuals are still alive for 10-12 minutes.

During this time, certain body movements and facial expressions make it clear that someone is drowning. Unfortunately, there are currently very few ways to prevent drowning. Children can wear floaties or life vests, but add the risk of making them flip upside down. Adult supervision is rare in the thousands of drowning cases yearly. It is truly surprising that for an activity as common as swimming and how often drowning occurs that no significant strides have been taken to ensure safety for all swimmers. A poolside, motorized Raspberry Pi-based monitoring system can be combined with machine learning to rapidly detect and prevent drowning in a majority of pool settings.



Alternate Hypothesis (H₁): The drowning detection system will accurately and rapidly recognize drowning in pools using visual and audial data and will alert for help.

Engineering Goals



Dynamic Hardware

The monitoring system will be able to monitor a wide horizontal and vertical area of the pool, both under and above water.



Maximize Detection Accuracy

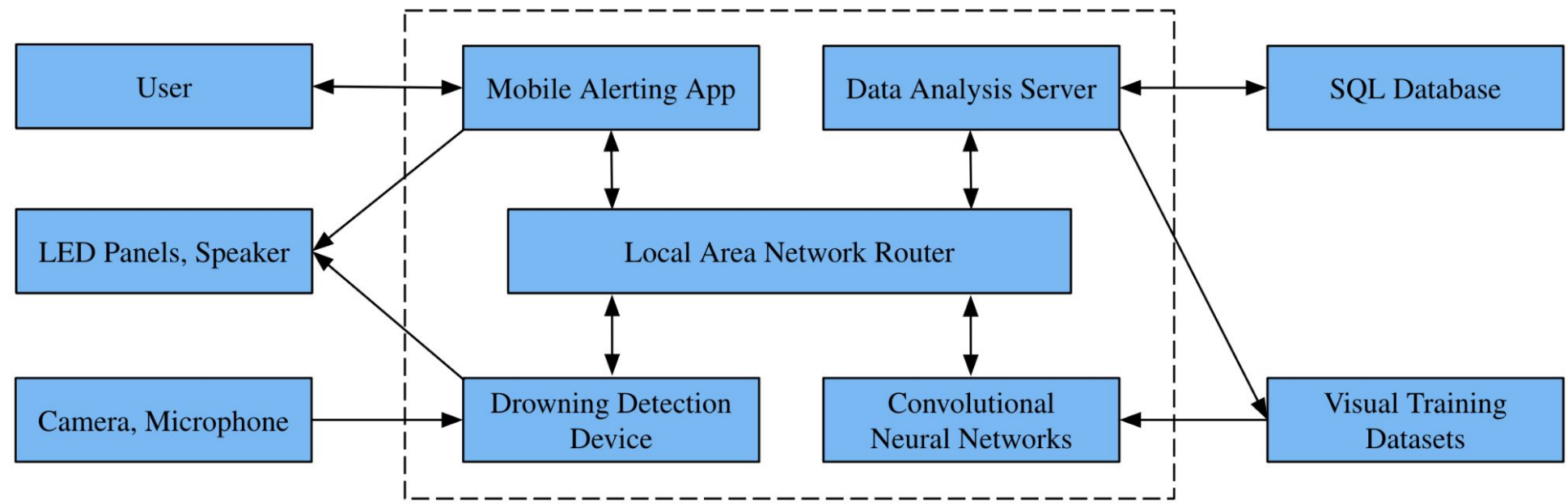
The machine learning component of the system will be designed to maximize accuracy and speed in drowning recognition.



Rapid Alerting Methods

An integrated mobile alerting app and peripheral speaker and light system will be used to send alarms signaling for help.

System Architecture



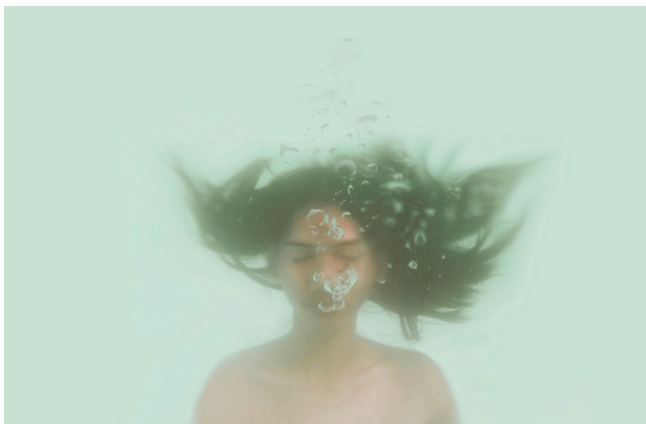
The constructed drowning detection system consists of four independent components: Drowning Detection Device, Data Analysis Server, Convolutional Neural Network, and the Mobile Alerting App. The Drowning Detection Device continuously streams visual and audial data to the Backend Server, which utilizes into a Convolutional Neural Network to assess the situation for possible drowning. Based on the machine learning prediction, peripheral alerting methods are updated and the mobile Alerting App is used to alert local officials and emergency contact numbers. All data transfer and communication is made through a user's Local Area Network (LAN) to connect through a central router to the specified internet protocol (IP) address of the Backend Server. All HTTP requests are sent over the WiFi for fast and high-capacity data communication.

Neural Network Training Data

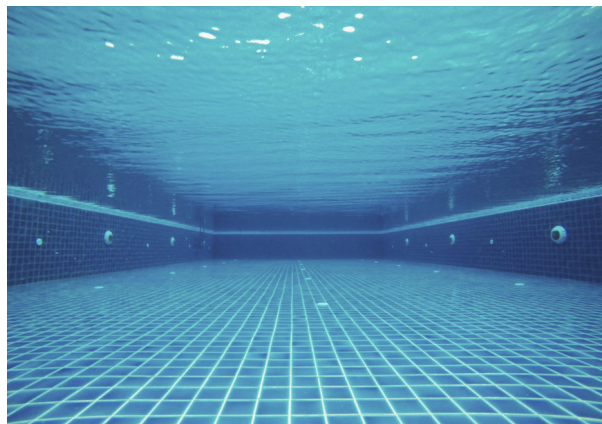
Swimming Image



Drowning Image



Empty Pool Image



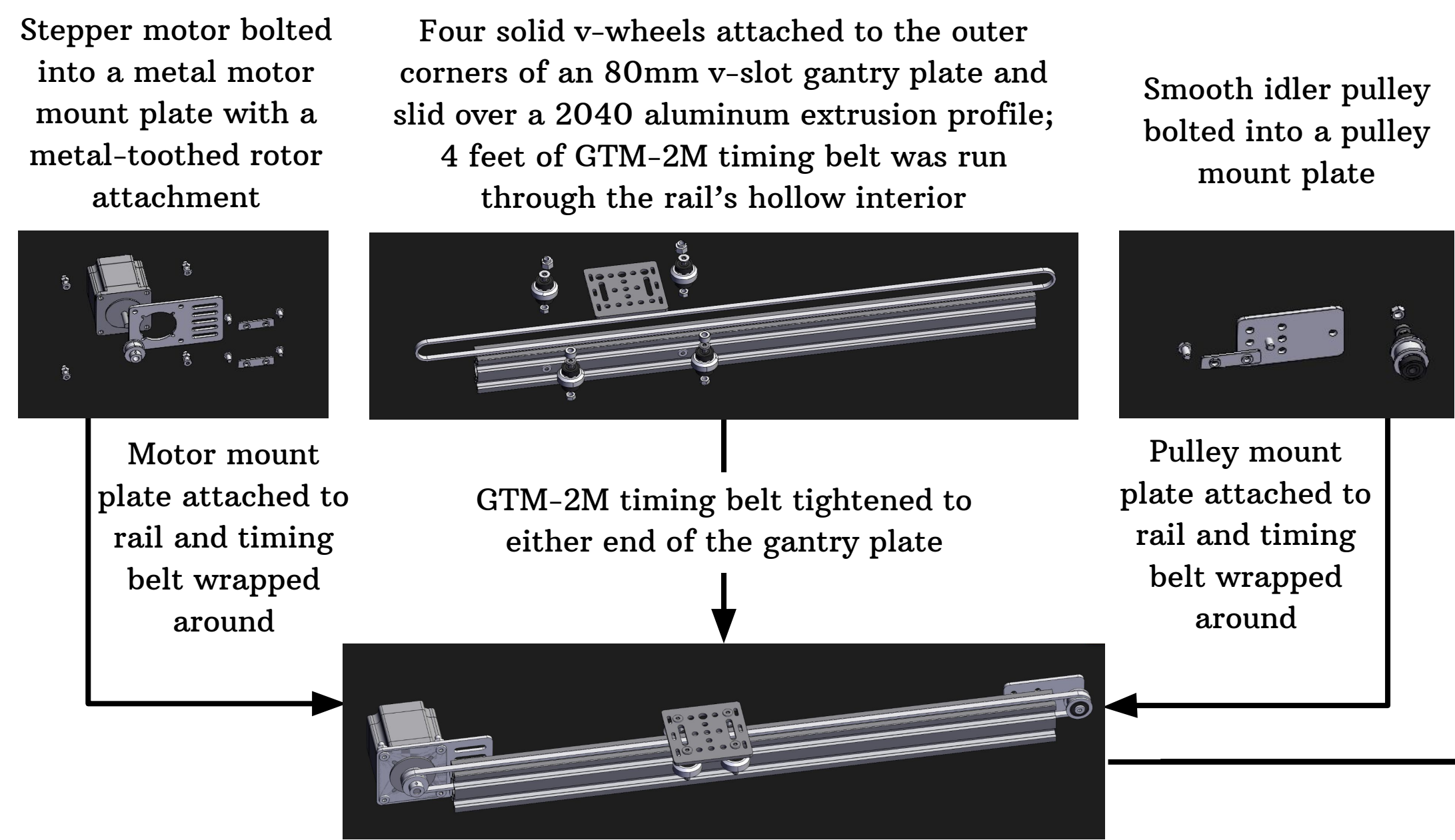
Because no visual datasets of swimming, drowning or empty pools currently exist, a custom dataset was created to train the CNN. Three primary classifications of training images were created for the following scenarios in pools: swimming, drowning, and empty pools. When assembling the dataset, above and below water, and daytime and nighttime images were taken. The composed dataset includes over 100 images of swimming, 100 images of drowning, and 50 images of empty pools - with instances at various heights, times, and ages of people. To further bolster the training dataset, the process of "augmentation" was implemented. Several copies of every images were made, and each copy was then randomly rotated, zoomed, stretched, or flipped. This makes the model more dynamic in recognizing various situations and prevents data overfitting.

Realtime Drowning Detection Using a Motorized Machine Learning Monitor System

Principal Component Analysis & Construction of Drowning Detection Device

1. Linear Actuator Construction

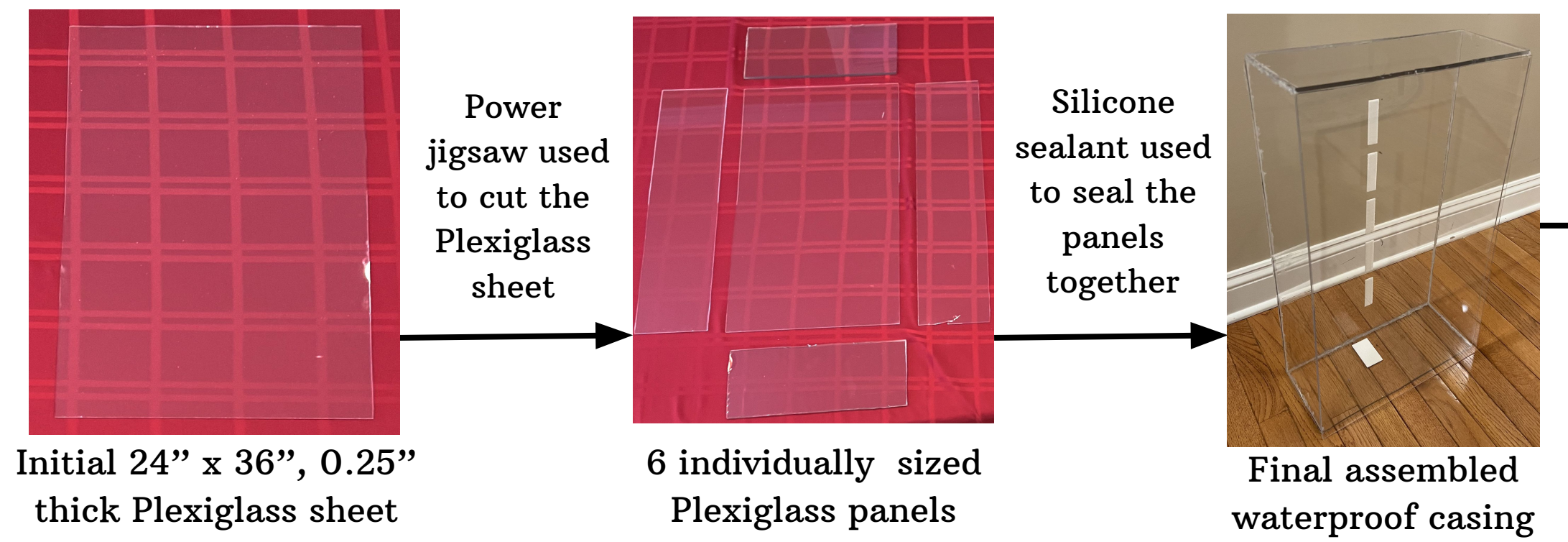
The first step of constructing the Drowning Detection Device was assembling the linear actuator. The function of the linear actuator is to translate up down and down with various modules and components attached to it. To do this, a 500mm 2040 aluminum extrusion profile, NEMA 17 stepper motor, solid v-wheels, smooth idler pulley, 80mm v-slot gantry plate, GT2-2M timing belt, metal mount plates, and various screws, nuts, spacers, cable ties were used.



The final constructed linear actuator utilizes the rotation of the stepper motor to push/pull the GTM-2M timing belt fastened to the gantry plate, which holds the servo motor, Logitech camera, and LED panels. On system startup, the gantry plate is reset to the bottom of the linear rail, and then begins translating above and below water.

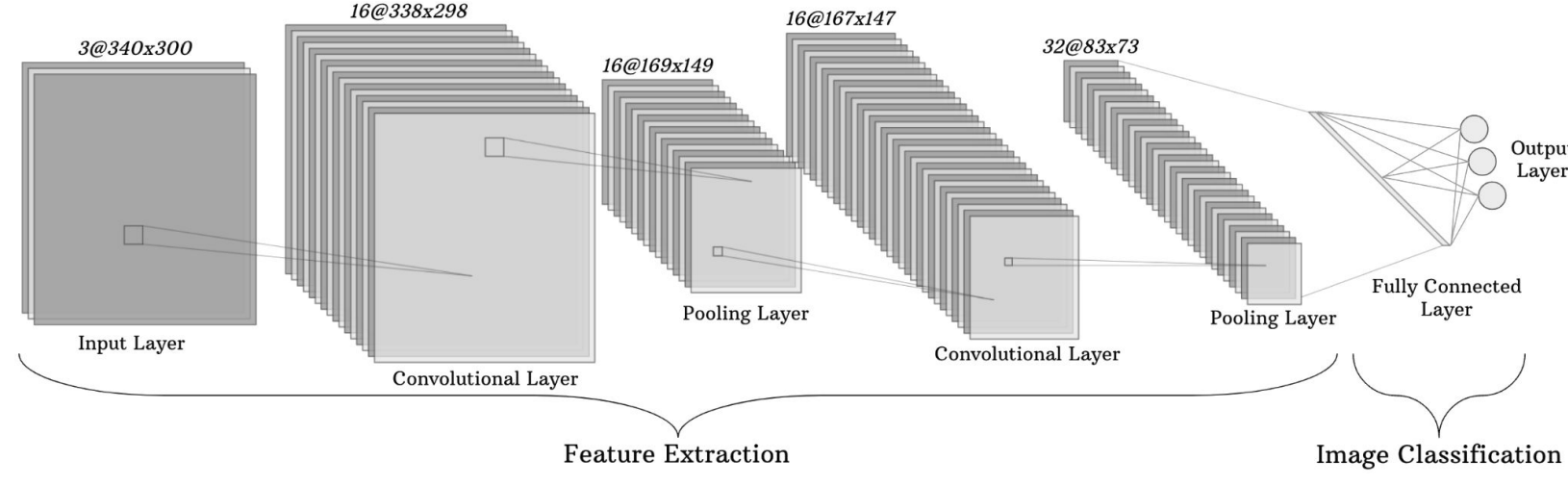
3. Plexiglass Case Construction

Because the Drowning Detection Device is semi-submerged and subject to constant pressure from the pool it is in, a durable and waterproof casing was constructed. Due to its durability, transparency, and ability to seal off outside fluids, Plexiglass was used as the primary material. The case was cut and assembled piece by piece, and allowed to dry before putting the mechanical and electrical hardware inside.

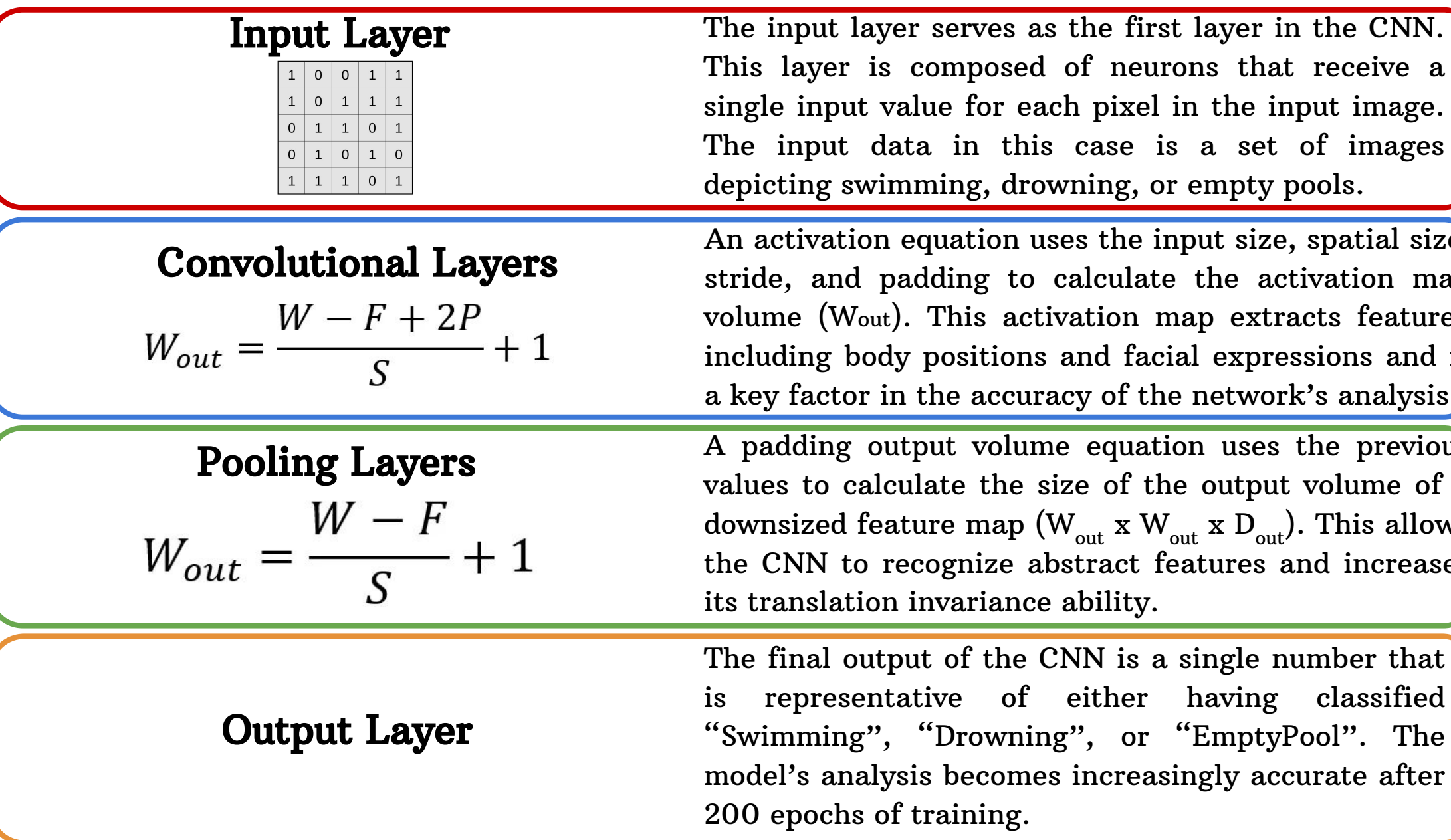


Machine Learning Overview

Convolutional Neural Network (CNN)

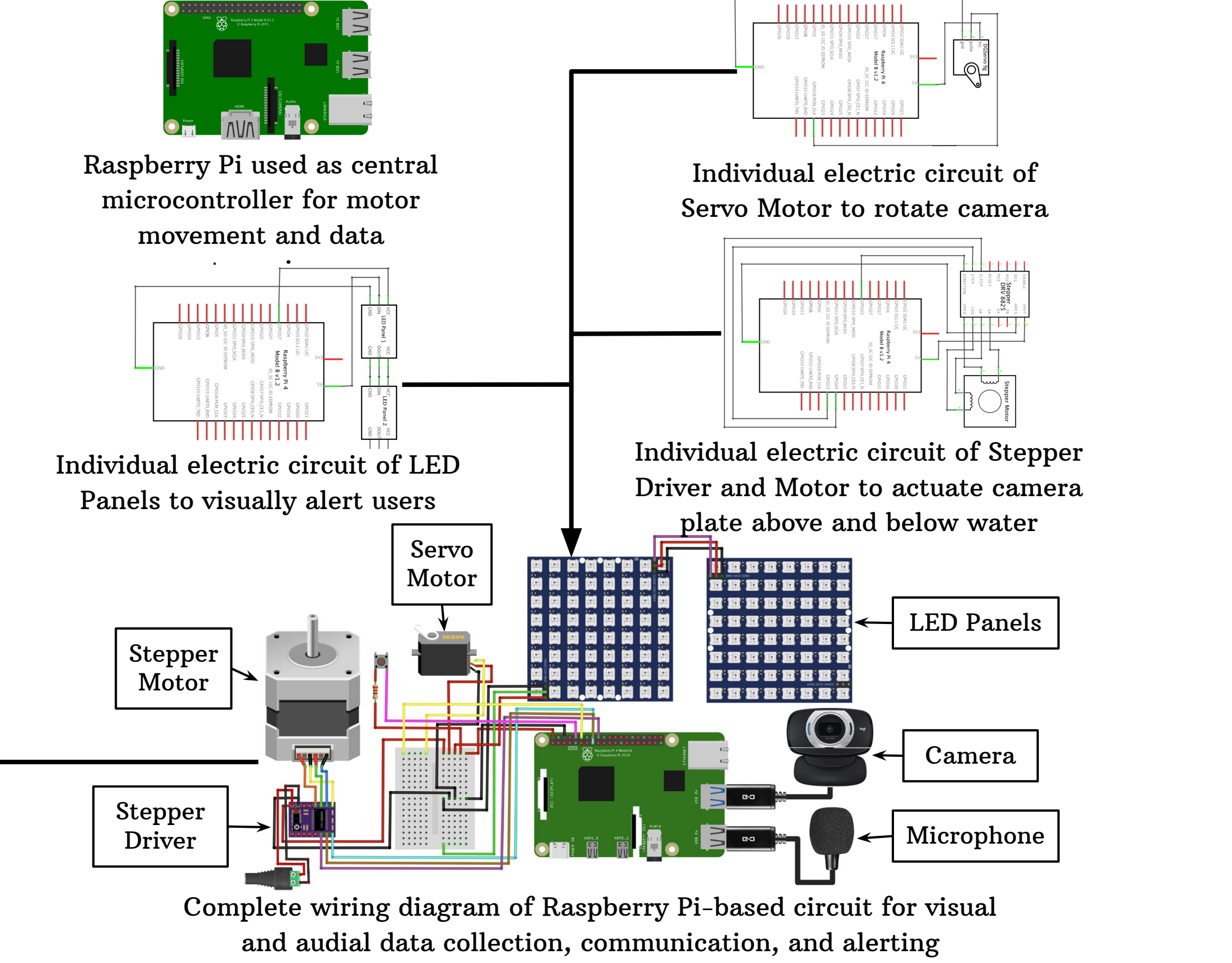


A convolutional neural network (CNN) was created to quickly and accurately recognize drowning from streamed visual data. It was programmed in Python using Google's TensorFlow machine learning library. The CNN works by passing a received image array through various convolutional layers which are designed to learn about specific features of the image. The network's parameters were fine tuned in correlation with the camera frame rate to allow for classification of an image before a new image is received. The Convolutional Neural Network was trained for 200 epoch iterations before turning on the system and is employed every time an image is sent from the Drowning Detection Device.



2. Raspberry Pi Circuit Construction

A Raspberry Pi 4 Model B with WiFi connectivity and the Linux-based Raspbian operating system is used to control the electrical modules and components used in the Drowning Detection Device. The Python programming language is used to program each component's functionality, which serve the collective purpose of streaming collected data to the backend server and providing sensory feedback.



4. Complete Drowning Detection Device

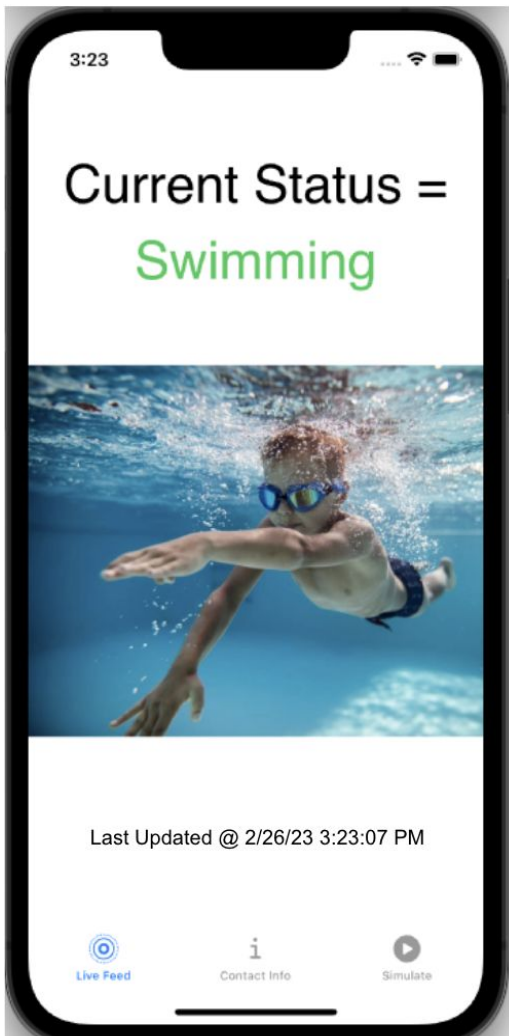


The Drowning Detection Device is the main component of the monitoring system and is used to detect and prevent drowning through its dynamic analysis and alerting methods. This waterproof device is secured to the side of a pool with its backside suction cups, and is half above and half below water. Its primary function is to capture a wide realtime visual and audial feed of the pool at various depths and angles. When an empty pool or swimming is detected, the Drowning Detection Device is on standby and slowly actuates above and below water, streaming data to the backend to make continuous predictions. As soon as drowning is detected, users are visually and audibly alerted.

Mobile Alerting App

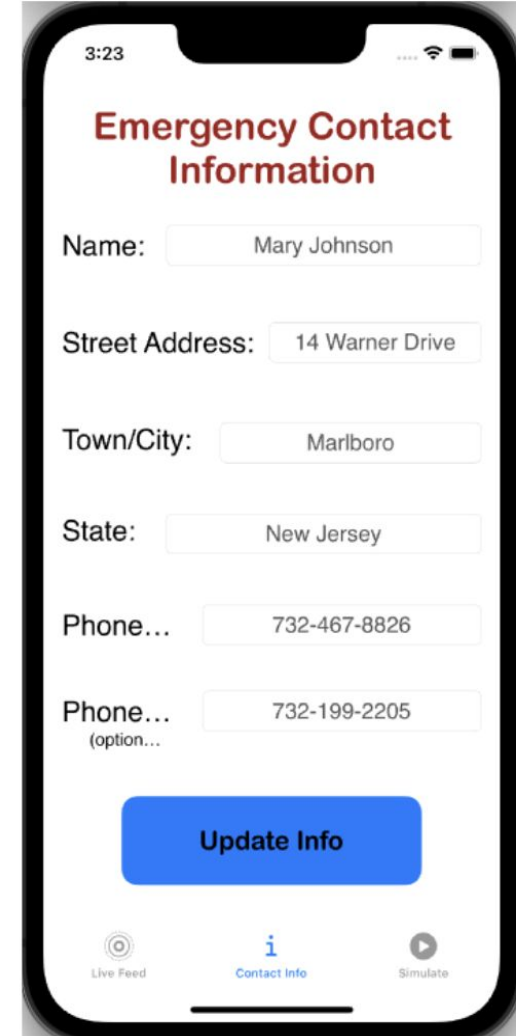
To create a seamless user experience with a live feed, an iOS mobile alerting application was constructed. The purpose of the user interface is to provide a live feed of the camera, give feedback on detected situations, and store emergency contact information. The mobile alerting app was constructed using Apple's Xcode development environment and the Swift programming language. The visual interface was designed first, including the user experience page flow, and images, labels, and buttons for user interaction. The programmatic functionality of the interface was designed next, including the communication sequence between the app and server. All interactions between the app and backend server are made via asynchronous REST API requests. The app consists of 3 unique pages: Live Feed, Contact Info, and Simulate.

Live Feed Page



The 'Live Feed' page shows the current detected status of the pool as well as a live view of the pool with realtime updating timestamps. It uses an asynchronous REST API requests to receive the live feed.

Contact Info Page



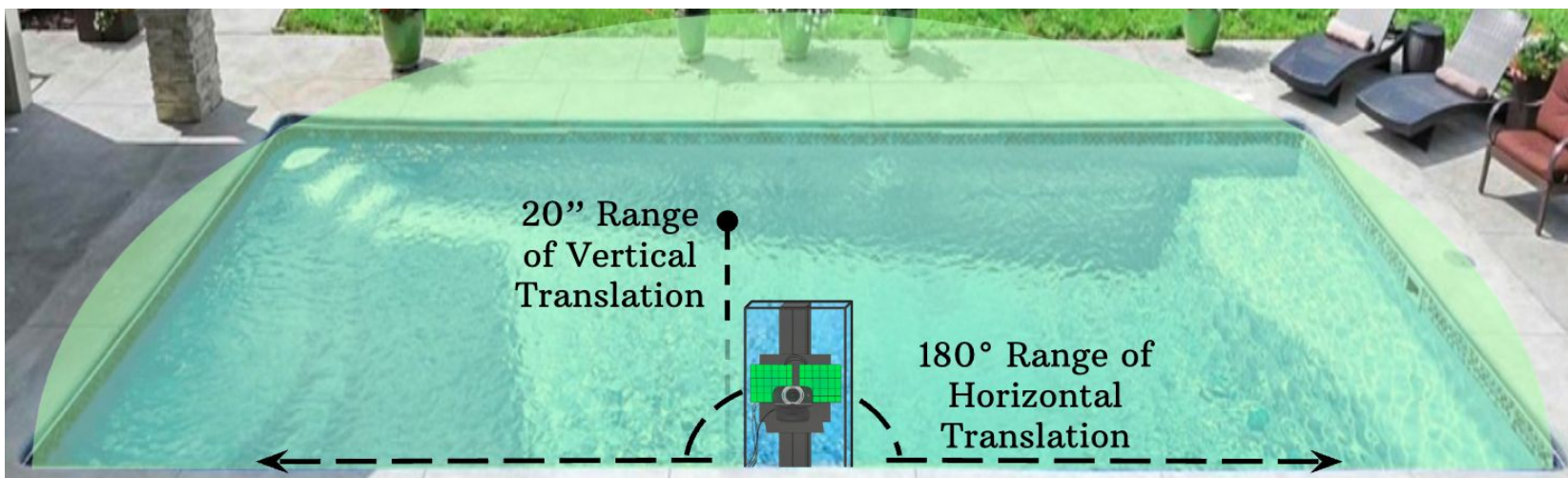
The 'Contact Info' page updates the homeowner/user's name, address, and phone numbers to reach out to and alert officials during potential drownings.

Simulate Page



The 'Simulate' page contains a button that triggers the drowning detection response of the monitoring system for testing purposes.

Range of Visibility Testing



To detect drowning in a wide variety of pool settings, the Drowning Detection Device contains a large range of mobility. The servo motor, which mounts the Logitech C615 camera, rotates 180°, while the linear actuator, which actuates the camera plate vertically, is 20" long. The hardware demonstrated capability of 20" vertical and 180° horizontal translation range paired with the camera's 81.5° field of view and high resolution accuracy of 95624.9 pixels per in² ensures the system can accurately monitor all areas of a pool in various sizes and shapes

Camera Resolution Accuracy - Derivation and Calculation

$$\text{Horizontal Angle of View (hV)} = \tan^{-1}(\frac{16}{9}) * \tan(78/2) * 2 = 69.5^\circ$$

$$\text{Vertical Angle of View (vA)} = \tan^{-1}(\frac{9}{16}) * (\tan(78/2)) * 2 = 42.5^\circ$$

$$\text{Diagonal Field of View} = \sqrt{(\text{hV})^2 + (\text{vA})^2} = \sqrt{(69.5^2 + 42.5^2)} = 81.5^\circ \text{ Field of View (per image)}$$

Use camera's horizontal and vertical angle of view to calculate the diagonal field of view.

$$\text{Pixels per in}^2 = \frac{(\text{camera resolution})}{(\text{cam width})(\text{cam height})} = \frac{(1920 * 1080)}{(2.8 \text{ in})(1.6 \text{ in})} = 462857.14 \text{ pixels/in}^2$$

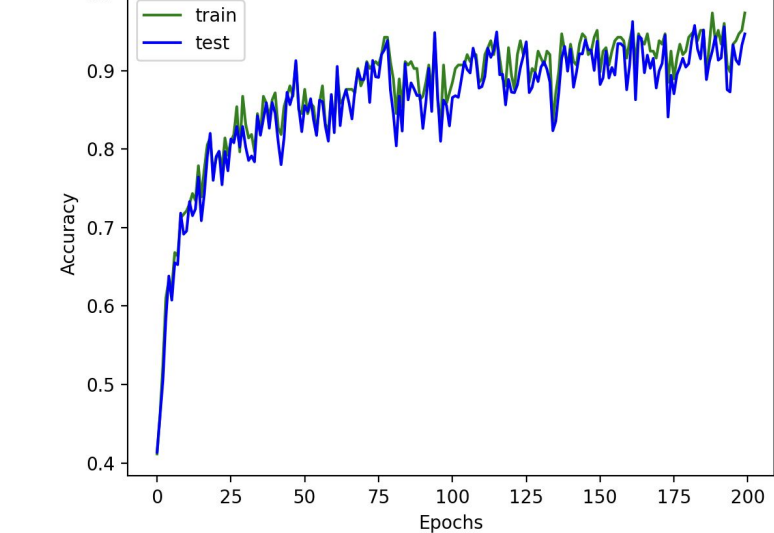
$$\text{Resolution Accuracy} = \frac{(\text{avg 1 year old size})(\text{pixels per in}^2)}{(\text{avg pool width})} = \frac{(29.75 * 462857.14)}{(144 \text{ inches})} = 95624.9 \text{ pixels/in}^2$$

Use camera's resolution specs to calculate average resolution accuracy for average conditions.

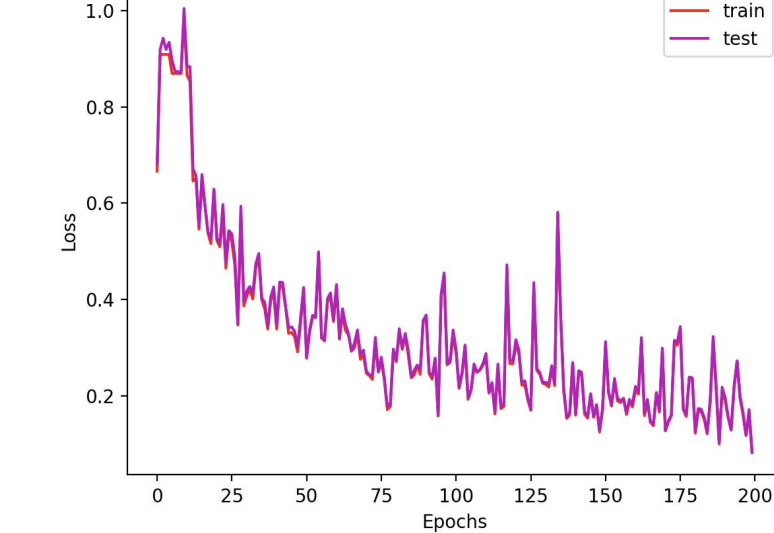
Neural Network Testing

Graphs displaying the loss and accuracy of the CNN model's training were computer generated using the Python 'matplotlib' library. A split of 70%-30% was applied to the training dataset in which 70% of images were used to train the model and the other 30% were used to validate the model's performance. The graphs depict the training and testing loss and accuracy of the model after each epoch iteration up to 200. The steadily decreasing loss and increasing accuracy represent the model's increasing success in image classification each epoch iteration. As shown in the progression graphs, training the model for 200 epoch iterations allows the model to maximize real-world prediction success without overfitting to the training data.

Drowning Detection Model Accuracy



Drowning Detection Model Loss



CNN Metrics

Metric Type	Metric	Final Value
Accuracy	Accuracy	0.9635
	Cosine Similarity	0.9747
Error	Loss	0.0817
	Mean Squared Error	0.0158

Findings and Implications

The proposed drowning detection system has the potential to revolutionize pool safety and prevent tragic drowning incidents, especially for children who swim without adult supervision. The system's yielded accuracy of 96.35% and loss of 8.17% achieved during testing on a custom visual dataset indicate that the machine learning model can successfully distinguish between swimming, drowning, and an empty pool. The incorporation of LED panels, speakers, and an iOS mobile alerting application also ensures rapid alerting methods so help can arrive quickly in the event of a drowning incident. The motorized Raspberry Pi-based monitoring system, with its dynamic and mobile hardware, is capable of capturing diverse images both above and below water, making it a comprehensive solution for pool safety.

The impact of this project is significant as it addresses a pressing issue that has resulted in thousands of deaths and injuries worldwide. By preventing drowning incidents, this system could potentially save many lives and prevent irreversible damage to the victim's health. The system's effective detection and alerting methods and ease of installation make it a feasible solution to reduce drownings on a larger scale in all pools.

Future Study

The proposed monitoring system has much potential to further increase its efficacy. One potential area for future study could be exploring the use of solar power or hydropower as the power source for the Drowning Detection Device. Solar panels could be used to charge a battery during the daytime while hydropower could be generated from the movement in the pool, making the system more eco-friendly and self-sustaining. Additionally, further advancements and diversity in training data could be made to increase the accuracy of the machine-learning model during different measurement conditions. Overall, new innovations and advancements in power supply options and detection ability can be implemented to create even an even more effective drowning detection system.

Finally, with the theoretical success of the hardware and neural network, real world testing of the system is planned for the 2023 summer. This is the last step before the drowning detection system can begin making a positive impact in residential pools.

References

<https://www.cdc.gov/drowning/facts/index.html#:~:text=Every%20year%20the%20United,22%20nonfatal%20drownings%20per%20day.>

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All Images, Figures, Graphs, etc. Created by the Researcher.

Additional References Found in Research Paper